Supplementary File S5: Experimental Protocol and Evidence Ledger

This supplement documents the latest available system-integration evidence and its limits. The runs below were performed against local virtual industrial endpoints. Evidence units are kept separate because they differ in task, actor, runtime configuration, and endpoint state. The associated machine-readable records and original CSV exports are supplied in `Supplementary-File-S5-Evidence.zip`. A companion HTML audit report provides an interactive step-by-step view.

I. Evidence Scope and Separation

The primary case is the completed 2026-10-31 document-driven HTTP/MQTT integration. It evaluates interface mapping, acquisition, read-only lead-task execution, and export consistency. The setpoint write/readback check was operator initiated and is reported separately from the agent task. Other model-backed, AgentTeam, protocol, and process-mission records are independent supporting evidence; counts and trajectories are not pooled. No physical plant was connected.

II. Document-Driven Virtual Integration

A. Configuration and execution

An interface contract declared six signals: HTTP enclosure temperature PV, setpoint, setpoint readback, and a 64-point thickness profile; and MQTT ambient temperature and humidity. The configuration created one new line, five DAQ nodes, one DCW node, a product, and a recipe. Provisioning reported 11/11 checks passed and zero cross-line write-point conflicts. The resulting Channel contained a lead and three configured worker roles. The archived activation evidence verifies only the lead as wired, and the retained task was assigned to that lead. The task was read-only and terminated `COMPLETED` at 100%.

B. Acquisition and export

The lead task queried five acquisition streams: four scalar signals and the vector profile. The task record contains 17 aggregated buckets for each scalar signal and 22 profile frames, each with 64 points. The observed ranges and export checks are summarized in Table I. Scalar query buckets and exported CSV rows have different aggregation semantics and must not be added together. The 19/19 export checks verify headers, manifest counts, value ranges, and vector shape. The CSV archive includes startup rows marked `offline`; thus, contract consistency does not imply that every row is a valid online observation.

TABLE I
Read-only task observations and exported data.

Signal	Task coverage	Observed range or structure
Enclosure PV	17 buckets	24.515–25.480 °C; mean 25.052 °C
SP readback	17 buckets	25.0 °C throughout the task window
Thickness profile	22 frames	64 points/frame; frame means 2.4975–2.5041 mm
Ambient temperature	17 buckets	20.5–23.44 °C
Ambient humidity	17 buckets	50.0–59.8%
Scalar CSVs	818 rows	Four files; includes startup offline rows
Vector CSV	3,456 rows	54 frames × 64 points/frame
Export validation	19/19	Manifest, headers, counts, ranges, and shape checks

C. Independent write/readback check

The virtual line began at a recipe setpoint of 25 °C. An early operator request for 27 °C was rejected because the 60-s write interval had not elapsed. A later operator/controller check wrote 27 and restored 25. A separate DAQ node observed both values. The HTTP write response echoed the request and is not treated as device readback. The write records have no task, approval, actor, or driver-event identifier linking them to the read-only agent task. This sequence therefore demonstrates an independent simulator write/readback path, not agent-controlled actuation.

III. Latest System Regression Results

Table II summarizes the 2026-10-31 T0–T4 acceptance report. These software and integration checks are not repeated process trials.

IV. Separate Agent, Route, and Protocol Evidence

Table III lists independent runs that are not combined with the primary integration case.

V. Four Scripted Process Missions

The September 28 mission records provide additional multi-scene simulator coverage. Each row is one deterministic trajectory; scripted routines, rather than an LLM, selected setpoints. These trajectories are not replications of the October document-driven integration and do not isolate causal action effects. Some terminal secondary variables were not archived.

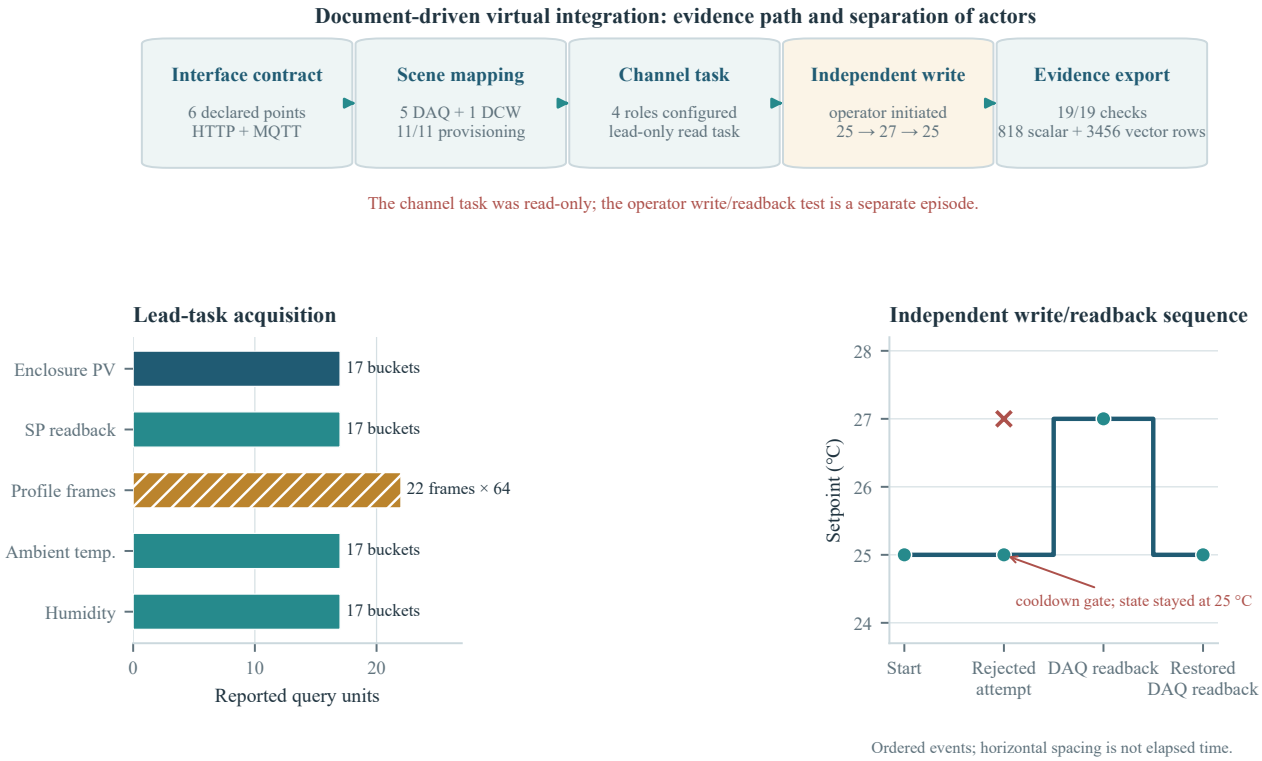


Fig. 1. Evidence sequence for the document-driven virtual-line case. The read-only lead task and operator-initiated writes are intentionally shown as separate paths. The lower panels summarize the retained scalar buckets, vector frames, and ordered setpoint observations. Horizontal spacing denotes event order only, not elapsed time or a causal trajectory.

TABLE II
LATEST SYSTEM REGRESSION CHECKS AND SCOPE.

Stage	Result	Qualification
T0 static	Lint passed after five existing lint errors were fixed; typecheck, documentation, localization, and README checks passed.	Completed after repair and rerun
T1 unit/contract	Unit: 915 tests, 913 passed, 0 failed, 2 skipped; MCP 14/14; plugin lifecycle 22/22; SDK surface passed; twin provider 5/5; skills 147/147.	Software checks only
T2 core integration	Line governance 44/44; document-driven virtual integration passed within the scope in Section II. High-load ontology, optimizer, UI, and model suites were not run in this campaign.	Limited coverage
T3 API-live	No qualified run: the script's fresh-instance registration precondition was unmet on the already-used service; other T3 suites were not run.	Not counted as pass or product failure
T4 plugin/tool surface	11 plugins with zero reported failures; 126/126 registered tools available.	Registry availability, not per-tool E2E

VI. Evidence Completeness and Manuscript Use

The primary integration case supports a bounded claim that a declared HTTP/MQTT virtual scene can be mapped into configured acquisition/control nodes, queried by a completed lead task, and exported with checked manifest/vector structure. The separate DAQ observations support one operator-initiated simulator write/readback demonstration. Software test counts support tested implementation paths. These data do not demonstrate multi-agent collaboration, approval-only causal effects, harness/model equivalence, repeated-run process performance, or physical deployment.

All source evidence used for the summary is copied without transformation where marked so in `SOURCE-MANIFEST.json`; original and delivered SHA-256 hashes are listed there and in `SHA256SUMS`. The original document-driven archive did not contain a contemporaneous checksum manifest; the package hashes establish integrity of the delivered copy, not a run-time-signed provenance chain. The audit report provides the step-by-step timeline, CSV row audit, and source-file hashes. The detailed machine records, run logs, exports, and correction to the earlier Qwen acceptance summary are included in the evidence ZIP.

TABLE III
SEPARATE EVIDENCE RECORDS AND THEIR ADMISSIBLE INTERPRETATION.

Run	Recorded result	Interpretation
OMP/Qwen, 2026-10-30	R3: setpoint stayed at 195 while a scripted approval was pending, then read 198 after approval. R4 wrote 200.5; the retained PV observation was 199.45. The simulator was already running.	One script-approved simulator write path and one later bounded write; not human-review efficacy or process improvement.
Qwen R2 query assertion	Original runner marked R2.1 pass; the task was `CANCELED`, and its deliverable values did not match the queried value. The assertion searched arbitrary numbers in full task history.	Query success is unverified and excluded. The earlier consolidated values 207.67/205.83 and 200.84 are not supported by the retained JSON.
OMP / USTC–DeepSeek-configured, 2026-10-31	Four tools were invoked and 50 DAQ points were recorded; acceptance was 10/13 and the task remained `WORKING`. No approval or successful write was retained. Runtime model attestation is absent.	Incomplete exploratory task; not an A2 model result.
P12 AgentTeam, 2026-10-30	Qualification passed; workflow and evidence-integrity gates failed. The task was canceled on a stale/future sample; no verified approvals or accepted writes; worker dispatch not verified.	Not a successful team episode.
Route fixture	Four deterministic simulator transitions with matching separate reads: cast-film 120 to 125/130 rpm and annealing 730 to 735/740 °C; synthetic administrator approval; no agent-selected action or post-write process response.	State-transition checks only.
Protocol matrix, 2026-10-29	Five local simulator protocols; 52 acquisition samples; four write-capable protocols; 24/24 out-of-window probes rejected; LLM disabled.	Deterministic protocol/guard checks only.
A1/A2	Approval-only ablation and full Harness×model matrix remain `NOT-RUN`; runner is not implemented.	No causal approval, harness, or model comparison is available.

TABLE IV
REPORTED PRIMARY VALUES IN THE FOUR SCRIPTED MISSIONS.

Scene	Target band; initial → reported final
Injection	32.15–32.85 g; 31.296 → 32.36 g
BOPET film	24.3–25.7 μm; 28.03 → 25.30 μm
Annealing	89–101 HV; 134.9 → 99.5 HV
Wastewater	2.8–4.0 mg/L DO; 0.194 → 3.54 mg/L